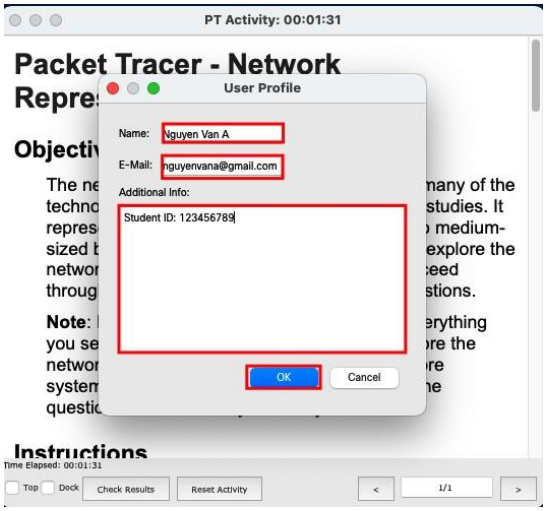
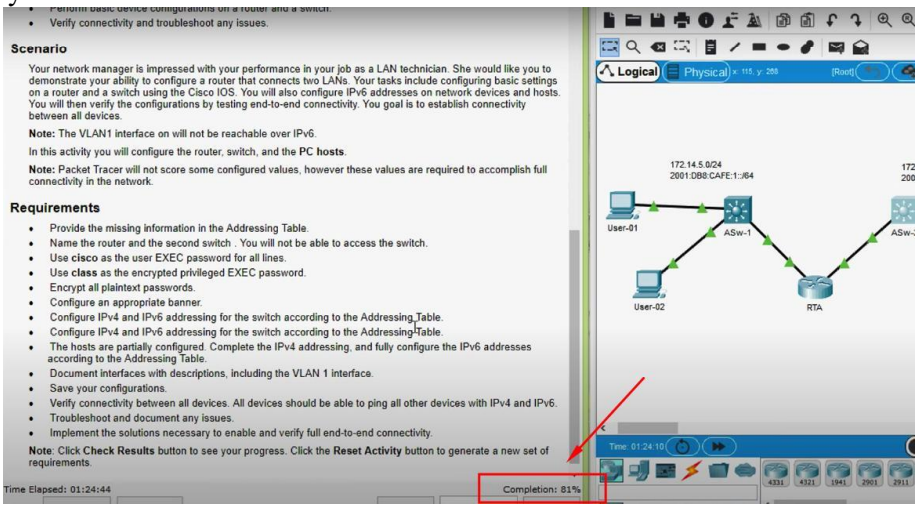
Networking Academy

Notes:

1. Remember to include your Name – Email – and Student ID inside the User Profile box



2. After finishing, please take a Packet Tracer screenshot which indicates the completion percentage as evidence for your work.



3. Please submit both .pdf and .pka on Blackboard before the deadline.

10.1.4 Packet Tracer - Configure Initial Router Settings

YouTube Link:
https://www.youtube.com/watch?v=bkHHZjWRd0c&list=PL8VT5seZ_GWhEZEDqq13KM3VuhzwpgZ9r&index=14&ab_channel=TechAcad

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Objectives

Part 1: Verify the Default Router Configuration

Part 2: Configure and Verify the Initial Router Configuration

Part 3: Save the Running Configuration File

Background

In this activity, you will perform basic router configuration tasks. You will secure access to the CLI and console port using encrypted and plain-text passwords. You will also configure messages for users who are logging into the router. These banners warn unauthorized users that access is prohibited. Finally, you will verify and save your running configuration.

Instructions Part 1: Verify the Default Router Configuration

Step 1: Establish a console connection to R1.

a. Choose a **Console** cable from the available connections.

b. Click **PCA** and select **RS 232**.

c. Click **R1** and select **Console**.

d. Click **PCA** > **Desktop** tab > **Terminal**.

e. Click **OK** and press **ENTER**. You are now able to configure **R1**.

Step 2: Enter privileged mode and examine the current configuration.

You can access all the router commands from privileged EXEC mode. However, because many of the privileged commands configure operating parameters, privileged access should be password-protected to prevent unauthorized use.

a. Enter privileged EXEC mode by entering the **enable** command.

```
Router> enable
Router#
```

Notice that the prompt changed in the configuration to reflect privileged EXEC mode.

b. Enter the **show running-config** command.

```
Router# show running-config
```

What is the router's hostname?

Router

How many Fast Ethernet interfaces does the Router have?

4 Fast Ethernet interfaces

How many Gigabit Ethernet interfaces does the Router have?

2 Gigabit Ethernet interfaces

How many Serial interfaces does the router have?

2 Serial interfaces

What is the range of values shown for the vty lines?

line vty 0 4

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c. Display the current contents of NVRAM.

```
Router# show startup-config startup-config is
not present
```

Why does the router respond with the **startup-config is not present** message?

Because there is no content in NVRAM whenever we do some configurations. Those configurations are saved in RAM

Part 2: Configure and Verify the Initial Router Configuration

To configure parameters on a router, you may be required to move between various configuration modes. Notice how the prompt changes as you navigate through the IOS configuration modes.

Step 1: Configure the initial settings on R1.

Note: If you have difficulty remembering the commands, refer to the content for this topic. The commands are the same as you configured on a switch.

a. Configure **R1** as the hostname.

b. Configure Message of the day text: **Unauthorized access is strictly prohibited.**

c. Encrypt all plain text passwords.

Use the following passwords:

1) Privileged EXEC, unencrypted: **cisco**
2) Privileged EXEC, encrypted: **itsasecret**
3) Console: **letmein**

Step 2: Verify the initial settings on R1.

a. Verify the initial settings by viewing the configuration for R1.

What command do you use?

show running-config

b. Exit the current console session until you see the following message:

```
R1 con0 is now available

Press RETURN to get started.
```

c. Press **ENTER**; you should see the following message:

```
Unauthorized access is strictly prohibited.

User Access Verification

Password:
```

Why should every router have a message-of-the-day (MOTD) banner? It s just a message to the intruders, we are giving a warning anyone like unauthorized access is strictly prohibited

If you are not prompted for a password before reaching the user EXEC prompt, what console line command did you forget to configure?

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We maybe forgot to give in command which we already given in console line

d. Enter the passwords necessary to return to privileged EXEC mode.

Why would the **enable secret** password allow access to the privileged EXEC mode and **the enable password** no longer be valid?

If we said these 2 passwords enable secret password will override the enable secret password hence to go to this privileged is its mode from user ignored we have to use secret password

If you configure any more passwords on the router, are they displayed in the configuration file as plain text or in encrypted form? Explain.

Once we give that service password encryption or the plaintext password will be encrypted also water password we are going to set in the future also will be encrypted

Part 3: Save the Running Configuration File

Step 1: Save the configuration file to NVRAM.

a. You have configured the initial settings for **R1**. Now back up the running configuration file to NVRAM to ensure that the changes made are not lost if the system is rebooted or loses power.

What command did you enter to save the configuration to NVRAM?

copy running-config startup-config

What is the shortest, unambiguous version of this command?

cop r st

Which command displays the contents of the NVRAM?

show running-config

d. Verify that all of the parameters configured are recorded. If not, analyze the output and determine which commands were not executed or were entered incorrectly. You can also click **Check Results** in the instruction window.

Step 2: Optional: Save the startup configuration file to flash.

Although you will be learning more about managing the flash storage in a router in later chapters, you may be interested to know that, as an added backup procedure, you can save your startup configuration file to flash. By default, the router still loads the startup configuration from NVRAM, but if NVRAM becomes corrupt, you can restore the startup configuration by copying it over from flash. Complete the following steps to save the startup configuration to flash.

a. Examine the contents of flash using the **show flash** command:

```
R1# show flash
```

How many files are currently stored in flash?

3 files

Which of these files would you guess is the IOS image?

3 33591768 c1900-universalk9-mz.SPA.151-4.M4.bin

Why do you think this file is the IOS image? Because we

can see the extension of this file is .bin

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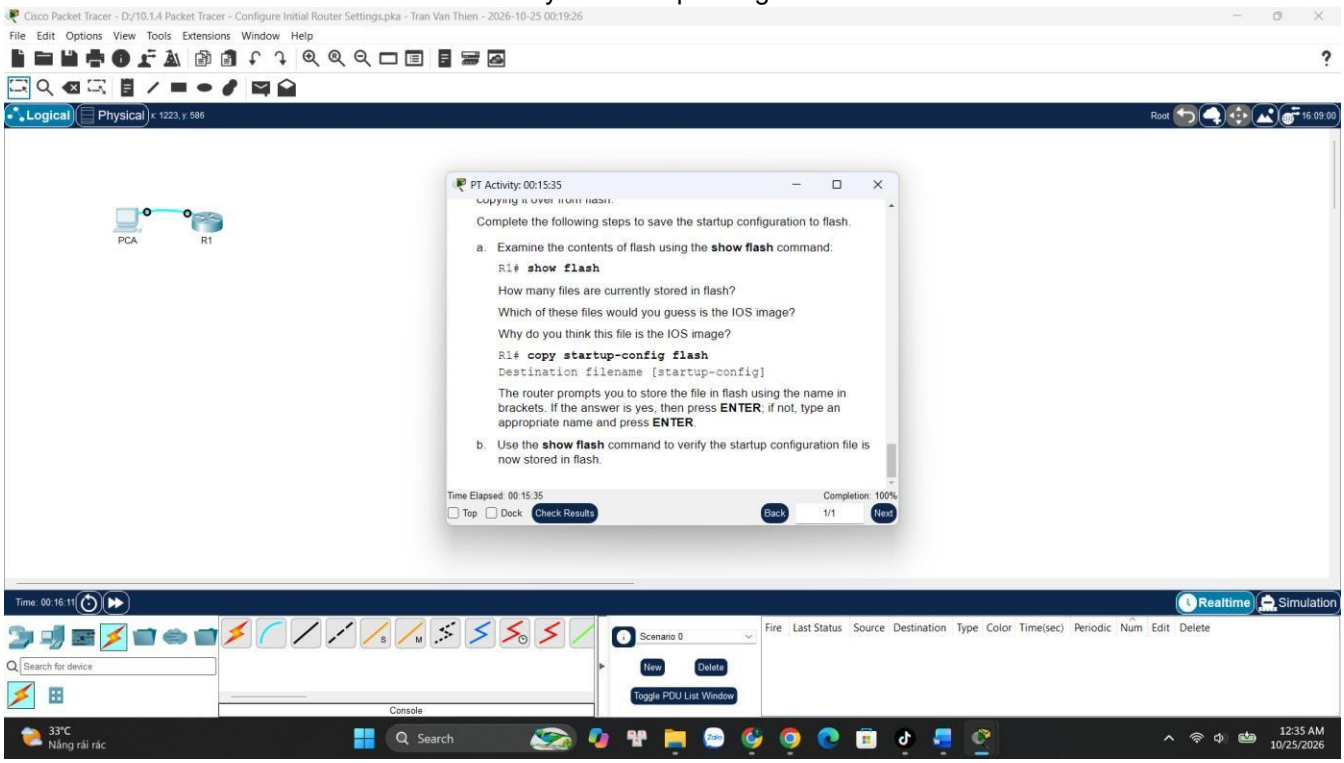
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b. Save the startup configuration file to flash using the following commands:

```
R1# copy startup-config flash
Destination filename [startup-config]
```

The router prompts you to store the file in flash using the name in brackets. If the answer is yes, then press **ENTER**; if not, type an appropriate name and press **ENTER**.

c. Use the **show flash** command to verify the startup configuration file is now stored in flash.



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